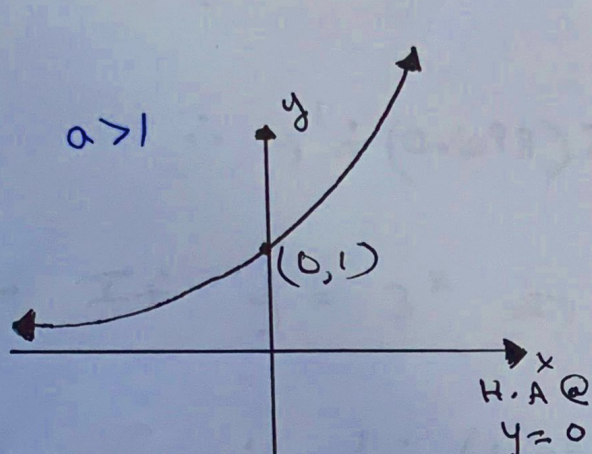


Exponential Functions

(1)

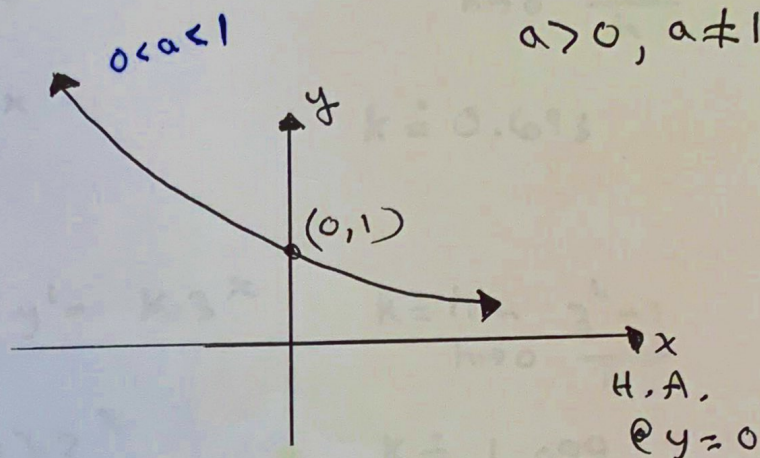
Exponential Function: $y = a^x$ $a \in \mathbb{R}$,
 $a > 0, a \neq 1$.



Increasing function

Domain: $(-\infty, \infty)$

Range: $(0, \infty)$



Decreasing function

Domain: $(-\infty, \infty)$

Range: $(0, \infty)$

NOTE: The derivative of an exponential function is an exponential function.

Given $f(x) = a^x$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h}$$

$$= a^x \cdot \lim_{h \rightarrow 0} \frac{a^h - 1}{h}$$

↑
exponential

↑ some constant

$a^x \cdot a^h$

Consider $y = a^x$

$$* \text{ If } y = 2^x \Rightarrow y' = k \cdot 2^x \quad k = \lim_{h \rightarrow 0} \frac{2^h - 1}{h}$$

$$\therefore y' \doteq (0.693) 2^x \quad k \doteq 0.693$$

$$* \text{ If } y = 3^x \Rightarrow y' = k \cdot 3^x \quad k = \lim_{h \rightarrow 0} \frac{3^h - 1}{h}$$

$$\therefore y' \doteq (1.099) 3^x \quad k \doteq 1.099$$

* Question:

Is there a number between 2 and 3

such that $\lim_{h \rightarrow 0} \frac{a^h - 1}{h} = 1$?

If so, we have found a function which is its own derivative.

→ Answer: yes!

" e " $\approx 2.71828 \dots$ (named after Euler "oiler")

$$\therefore \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = 1$$

$$\therefore \text{ If } y = e^x \Rightarrow y' = e^x$$

In general, if

$$f(x) = e^{g(x)}, \text{ then } f'(x) = e^{g(x)} \cdot g'(x)$$

example 1: Differentiate the following.

a) $y = 2e^x$

$$y' = 2e^x$$

b) $y = e^{2x}$

$$y' = e^{2x} \cdot (2x)' = e^{2x} \cdot 2$$

$$y' = 2 \cdot e^{2x}$$

c) $y = e^{-3x^2}$

$$y' = e^{-3x^2} \cdot (-3x^2)'$$

$$y' = -6x \cdot e^{-3x^2}$$

C.#

x=0

d) $f(x) = 3x^2 \cdot e^x$

$$f'(x) = 6x \cdot e^x + 3x^2 \cdot e^x$$

$$f'(x) = 3x e^x (2 + x)$$

$$3x(2+x)e^x$$

C.#

$$x=0, -2$$

e) $f(x) = \frac{x^2}{e^{2x+1}}$

$$f'(x) = \frac{2x e^{2x+1} - x^2 \cdot e^{2x+1} \cdot (2)}{(e^{2x+1})^2}$$

$$f'(x) = \frac{2x e^{2x+1} (1-x)}{(e^{2x+1})^2}$$

$$= \frac{2x(1-x)}{e^{2x+1}}$$

crit #
0, 1

f) $f(x) = e^{x^2} \cdot \sin(x^2) \leftarrow [0, 2\pi]$

(4)

$$f'(x) = (e^{x^2})' \sin(x^2) + e^{x^2} (\sin(x^2))'$$

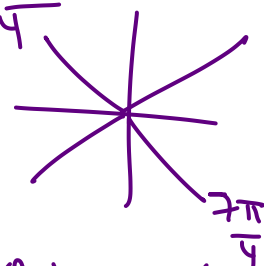
$$f'(x) = 2xe^{x^2} \sin(x^2) + e^{x^2} \cos(x^2) \cdot 2x$$

$$f'(x) = 2xe^{x^2} (\sin(x^2) + \cos(x^2))$$

crit # : 0

$$\sin(0) + \cos(0) = 0$$

$$\sin(0) = -\cos(0)$$



g) $f(x) = \pi e^{\pi} \sin(e^x) + e^{\pi}$

$$-\pi e^{\pi-1} \tan(0) = -1$$

$$0 = \tan^{-1}(-1)$$

$$f'(x) = \pi e^{\pi} \cos(e^x) \cdot e^x + 0$$

$$0 = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$f'(x) = \pi \cdot e^{x+\pi} \cdot \cos(e^x)$$

$$x^2 = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$x = \pm \sqrt{\frac{3\pi}{4}}, \pm \sqrt{\frac{7\pi}{4}}$$

h) Determine the equation of the tangent

to the curve $y = e^{2x-1}$ at $x=1$.

$$y = e^{2x-1}$$

$$y' = 2e^{2x-1}$$

$$m_t = 2e \Big|_{x=1}$$

$$y - y_1 = m(x - x_1)$$

$$y - e = 2e(x - 1)$$

slope-y-int

$$y = 2ex - e$$

leave exact

$$Ax + By + C = 0$$

$\times A, B, C$ integers $\left. \begin{array}{l} 2ex - y - e = 0 \\ A > 0 \end{array} \right\}$ not standard form.

$$y = e^{2(1)-1}$$

$$y = e$$

$$(1, e)$$

y-int